

Real-Time Streaming Data Pipeline Specification

1. Messaging Infrastructure

Apache Kafka cluster runs across 9 broker nodes with a replication factor of 3 for critical event topics.

Average event ingestion throughput sustains 450,000 messages per second during peak hours.

Deduplication & Schema Evolution

2. Stream Processing

Apache Flink handles 10-minute sliding event-time deduplication with exactly-once state semantics.

Confluent Schema Registry enforces Avro backward-transitive schema compatibility.